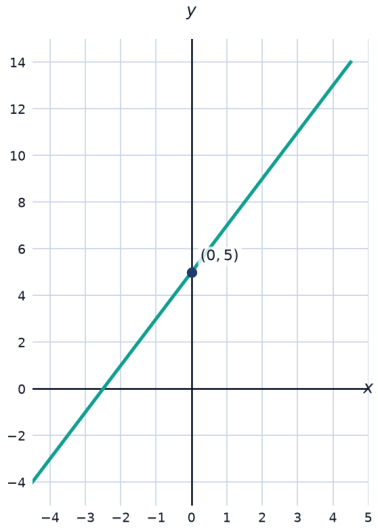
Functions & Transformations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

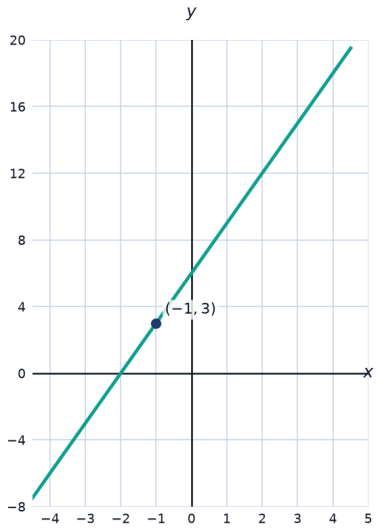
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 5$ in transformations of y=x, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 6$ in slope-intercept form, evaluate $f(2)$.



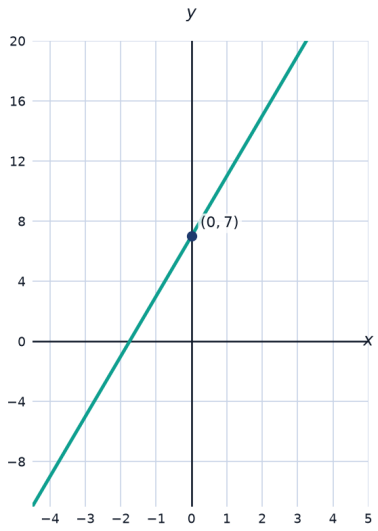
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for slope-intercept form consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 4x + 7$ in standard form, evaluate $f(2)$.

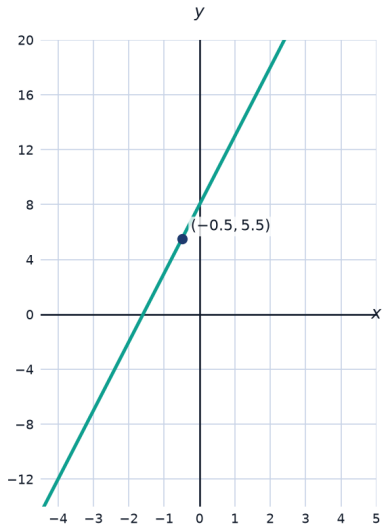


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for standard form consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 5x + 8$ in point-slope form, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for point-slope form consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let $f(x) = \frac{2x-7}{5}$. (a) Find $f^{-1}(x)$. (b) Solve the equation $f(f^{-1}(x)) = 9$. (c) Find the value of k for which $f(k) = f^{-1}(k)$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Determine the maximal real domain of the function $f(x) = \frac{\sqrt{3x-12}}{x^2-25}$. Express your answer in interval notation.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle OAB, $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$. Point P lies on AB such that $AP : PB = 1 : 2$. Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of $y = f(x)$ passes through the points $(-2, 8)$ and $(4, -6)$. If $g(x) = 3f(x-5) + 2$, determine the corresponding points on the graph of $g(x)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Radicals & Exponents

Given $f(x) = x^2 - 4x + 7$ with restricted domain $x \geq 2$, find an explicit formula for $f^{-1}(x)$ and state its domain.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Linear Equations & Lines

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Rational Expressions

Section I of EST I Math contains 20 questions in 25 minutes. If a student answers the first 12 questions in an average of 50 seconds each, how many seconds per question are available for the remaining 8 questions?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Statistics & Data Analysis

A student has scores of 78, 85, 92, and 81 on four exams. What score must the student earn on the fifth exam to achieve an overall arithmetic mean of 85 across all five exams?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratic Functions

Level 4 · SynthesisLine ℓ passes through $(2, -1)$ and is perpendicular to the line $3x + 6y = 18$. If line ℓ also passes through $(k, 9)$, what is the value of k?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Exponents & Polynomials

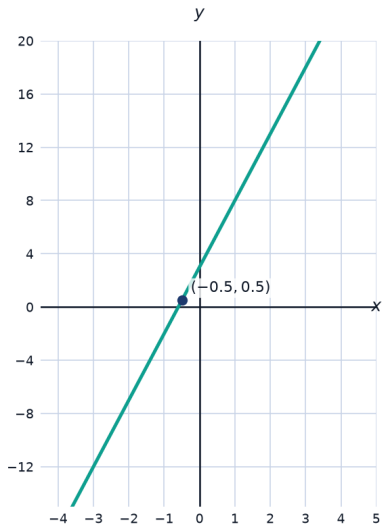
Level 1 · FoundationsThe Digital SAT Math section allocates 70 minutes across two modules of equal length. If each module contains 22 questions, what is the exact average time available per question in seconds?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 3$ in review: slope as rate of change, evaluate $f(2)$.



a. 13

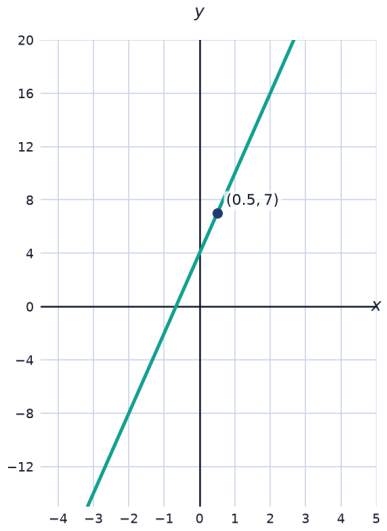
b. 18

c. 10

d. 17

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 4$ in the slope formula, evaluate $f(2)$.



a. 16

b. 22

c. 12

d. 20

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: slope as rate of change consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for the slope formula consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arithmetic sequence has first term -9 and common difference -3. Find the 10th term.

a. -33

b. -37

c. -36

d. -39

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Graph the solution of $x + y \leq 4$ and $x - y \leq 2$. Which point is in the solution region?

a. $(0, 5)$

b. $(3, 2)$

c. $(1, 1)$

d. $(5, 0)$

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student needs at least an 85 average across 4 exams. If the first three scores are 82, 88, and 79, what is the minimum score required on the fourth exam?

a. 89

b. 91

c. 93

d. 95

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A researcher models exam scores with the line of best fit $y = 5.2x + 58$, where x is the number of hours studied and y is the predicted score. A student who studied 5 hours earned a score of 88. What is the residual (actual score minus predicted score) for this student?

a. +4

b. -4

c. +6

d. +8

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:

a. No conclusion possible

b. There exists $c \in (1, 4)$ with $f(c) = 0$

c. $f(2) = 1$

d. f is constant

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = -2f(x + 1) - 3$ applies which transformations to $y = f(x)$?

a. Left 1, stretch 2, down 3 (no reflect)

b. Right 1, reflect, stretch 2, up 3

c. Right 1, stretch 2, down 3

d. Left 1, reflect over x-axis, vertical stretch 2, down 3

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31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The Intermediate Value Theorem states that if $P(a)$ and $P(b)$ have opposite signs, then:

a. There are two roots

b. There is a root between a and b

c. $P(0) = 0$

d. $P(a)P(b) = 0$

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

On a multiple-choice question with 4 answer choices, what is the probability of guessing the correct answer at random?

a. 0.20

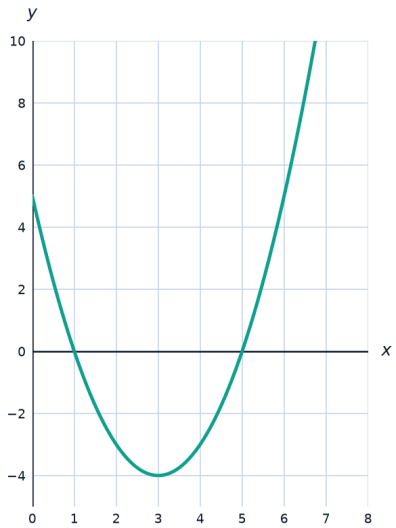
b. 0.25

c. 0.33

d. 0.50

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A function is defined by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ The graph of $y = f(x)$ is a parabola. A straight line with equation $y = k$ intersects the graph of $y = f(x)$ at exactly one point. \\ Find the possible value(s) of k.



a. $k = -4$

b. $k = 5$

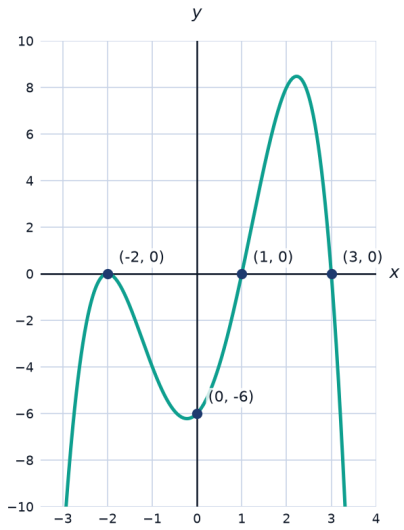
c. $k = -4$ or $k = 5$

d. $k \geq 5$

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?



a. 24

b. 18

c. 17

d. 14

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35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?



a. $f(x) = (x + 2)(x - 1)(x - 3)$

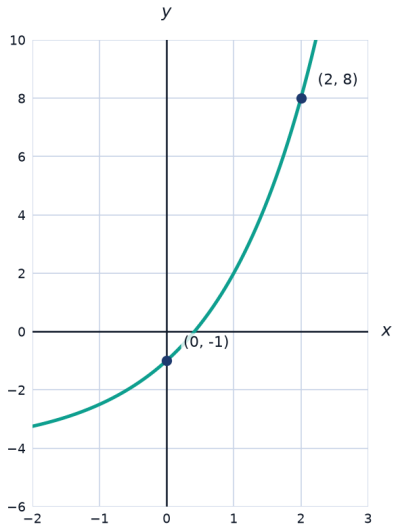
b. $f(x) = (x - 2)(x + 1)(x + 3)$

c. $f(x) = -(x + 2)(x - 1)(x - 3)$

d. $f(x) = (x + 2)(x - 1)^2(x - 3)$

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?



a. $-\frac{5}{2}$

b. $-\frac{7}{2}$

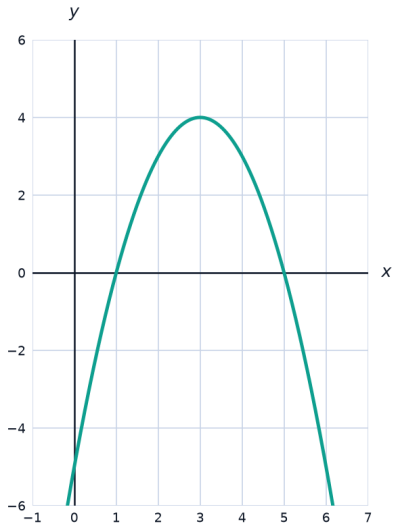
c. $-\frac{13}{4}$

d. $\frac{3}{2}$

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37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?



a. -5

b. 3

c. 4

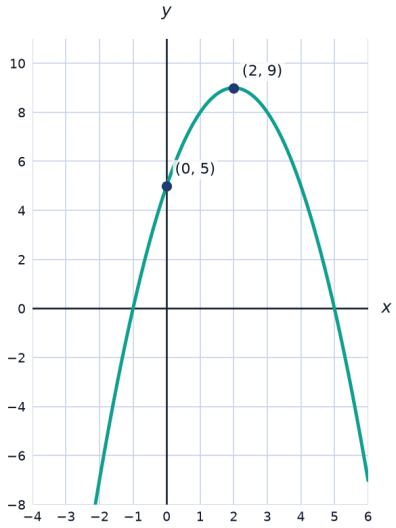
d. 7

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Statistics & Data Analysis

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?



a. -4

b. -2

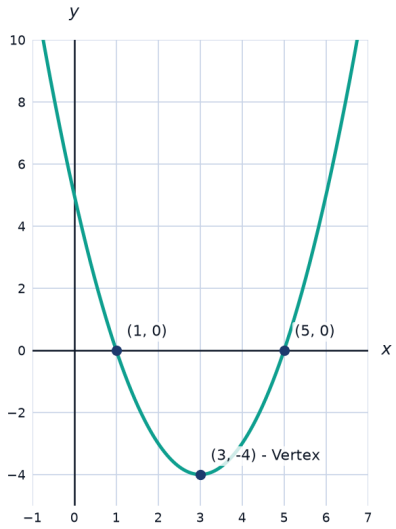
c. -1

d. 1

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratic Functions

The graph of the function $y = ax^2 + bx + c$ is shown below. The function passes through the points $(1, 0)$ and $(5, 0)$, and its vertex has a y-coordinate of -4. Determine the value of b.



a. -6

b. 6

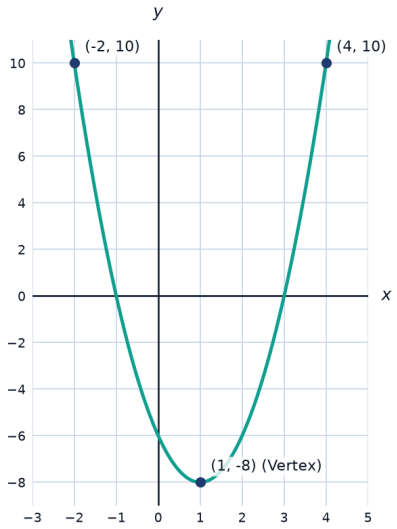
c. -4

d. 4

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Exponents & Polynomials

The graph of the quadratic function $g(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-2, 10)$ and $(4, 10)$, and its vertex is at $(1, -8)$. Determine the value of a.



a. 1

b. 2

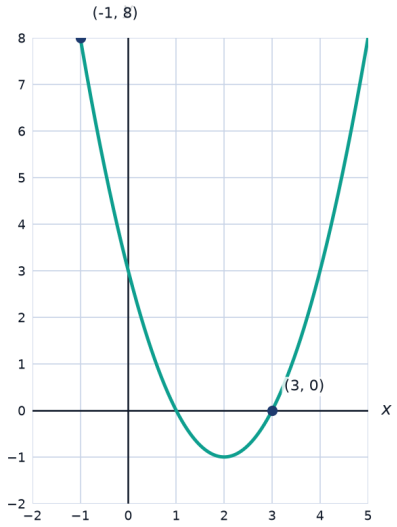
c. -1

d. -2

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.



a. $a = -2, b = 3$

b. $a = 2, b = -3$

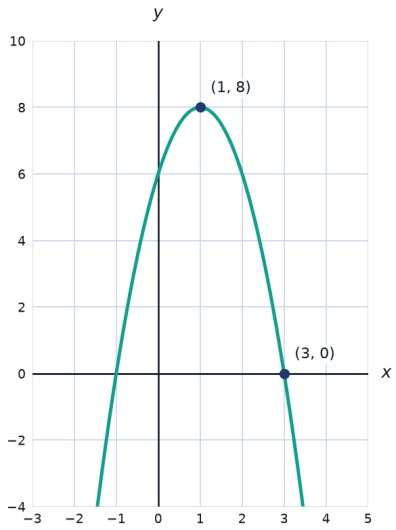
c. $a = -4, b = 1$

d. $a = 4, b = -1$

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Systems & Inequalities

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?



a. -10

b. -6

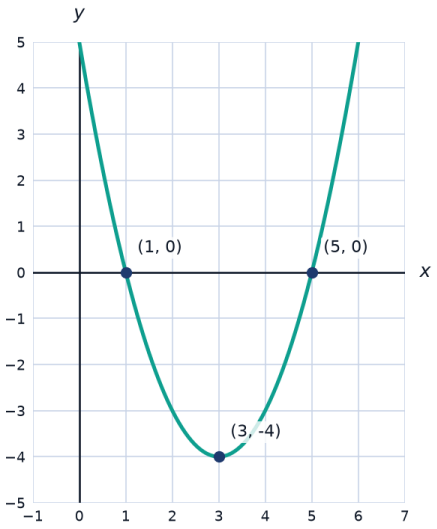
c. 6

d. 10

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Sequences & Series

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(1, 0)$ and $(5, 0)$. If the vertex of the parabola is at $(3, -4)$, what is the value of a?



a. 1

b. -1

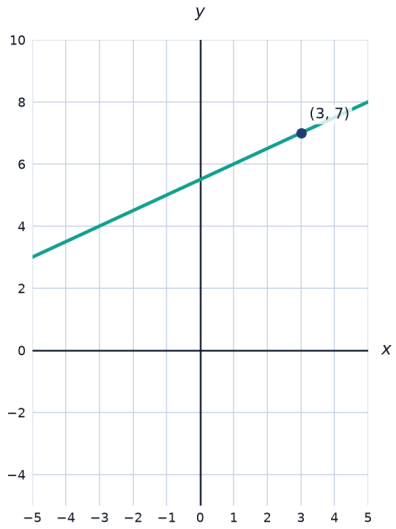
c. 2

d. -2

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Factoring Polynomials

The graph of the linear function $y = mx + b$ is shown below. If the graph passes through the point $(3, 7)$ and the slope m is $\frac{1}{2}$, what is the value of b?



a. $\frac{1}{2}$

b. $\frac{11}{2}$

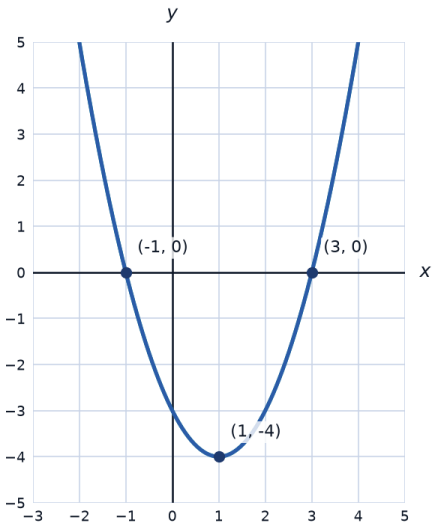
c. $\frac{13}{2}$

d. $-\frac{1}{2}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Radicals & Exponents

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(-1, 0)$ and $(3, 0)$. The vertex of the parabola is at $(1, -4)$. Determine the value of a.



a. -1

b. 1

c. 2

d. -2

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